Late cosmic acceleration as a reading of the observer's clock tied to the growth of structure: registered tests against DESI DR2, supernovae, CMB distance priors and growth data

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Abstract

In its own combinations with the CMB and supernovae, DESI DR2 prefers an evolving dark energy with $w_0 > -1$ and $w_a < 0$ over ΛCDM at 2.8–4.2σ (DESI Collaboration 2025). This record tests the dark-energy history of the author's framework, Mathematical Materiality (MM), against DESI DR2 BAO, compressed CMB distance priors (Chen, Huang and Wang 2019) and three supernova samples, and against the growth of structure. It then proposes a phenomenological hypothesis, read with MM by coherence: the late acceleration as a reading of the observer's clock that changes with the growth of cosmic structure. Every test was registered with SHA-256 fingerprints before its computation (knowing earlier results on the same data, as each registration states); additions decided after seeing results are labelled; failed predictions are reported. An independent AI system checked each block in one to nine rounds. It is an exploratory, preliminary study.

Can the acceleration of the universe be an effect of the observer's clock? Registered tests with DESI DR2, supernovae, background radiation and growth of structures

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The archived text is in English.

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Before starting: the words you need

The universe expands. Distant galaxies move away from us. The *expansion rate*, H, says how fast this happens. It is measured in km/s/Mpc: kilometres per second for each megaparsec of distance. A megaparsec is a huge distance, more than three million light years.

The late acceleration. For a few billion years the expansion has not been slowing down: it has been accelerating. The most used explanation is a *dark energy*, a substance that pushes the expansion.

ΛCDM. It is the standard model of cosmology. It uses a *cosmological constant*, Λ: a dark energy that never changes. It also contains cold dark matter (CDM).

w, the equation of state. For a substance, w is the pressure divided by the energy density. For the cosmological constant w = −1, always. If w changes over time, dark energy *evolves*.

The CPL model. It is the most used way to write a w that changes: w(a) = w0 + wa (1 − a). Here a is the *scale factor*: it is 1 today and it was smaller in the past. w0 is today's value; wa says how much w changes.

Redshift, z. The light of distant galaxies is stretched during its journey, because the universe expands. This stretching is written z. The larger z is, the farther away and the further back in time we look.

The data used for the expansion.

- **DESI DR2** measures the *baryon acoustic oscillations* (BAO). They are a typical distance imprinted in the distribution of galaxies: they act as a cosmic ruler.
- **The cosmic microwave background** (CMB) is the oldest light of the universe. Here it enters with three summary numbers, called *distance priors*.
- **Supernovae** are stars that explode with a known luminosity. Their distance is obtained from their light. Here three samples are used: Pantheon+, Union3 and DES-Dovekie. Another sample, DES-SN5YR, serves only as a check.

For the growth of structures other data are used: SDSS, DESI DR1 and 6dFGS (section 8).

The growth of structures. At the beginning the universe was almost uniform. Then gravity gathered matter into galaxies and clusters. The *linear growth factor*, D, says how much the small differences in density have grown. σ8 measures how *lumpy* matter is today, on spheres of 8 h^{-1} Mpc. fσ8 says how fast structures grow.

The χ^2 and the $\Delta\chi^2$. The χ^2 (read *chi-squared*) is an error score: the lower it is, the better a model follows the data. $\Delta\chi^2$ is the difference from ΛCDM. A negative $\Delta\chi^2$ means that the model does better than ΛCDM.

σ. It is the typical error. A result *at 3σ* is three typical errors away from another.

Registering. A test is *registered* when it is written down and dated before doing the calculations.

The translations. The author writes his ideas in words, as *premises*. To do the calculations they must be *translated* into formulas. The translations are choices of whoever does the calculations. A negative outcome hits a translation, not the premise.

At a glance

The question. DESI (2025) combined its data with the CMB and with supernovae. In these combinations the CPL model, with an evolving dark energy, is preferred over ΛCDM, the standard model with a cosmological constant (w = −1). Can the late acceleration be read in another way? Can it be an effect of how time is read by an observer who lives inside growing structures, instead of an extra substance?

The idea in a few lines.

- We take a universe of matter and radiation only, without dark energy. It is the *bare* universe.
- The expansion rate *read by the observer* is higher than the bare one by a factor 1 + δ0 S(z).
- S(z) is a *tracer*, that is, a measure, of the growth of cosmic structures in the whole universe. It is 1 today and tends to 0 if there are no structures.
- Mostly S = [D(z)/D(0)]^n, where D is the linear growth factor.
- The clock model is an expansion history different from that of the published MM 2.32. A way of translating the counts of 2.32 into this clock law has not yet been built.

An example from everyday life. If your watch runs slow without your knowing it, everything you measure seems faster to you. Here the idea is similar: the clock of those who are inside structures makes the expansion seem faster.

The data. The DESI DR2 BAO, the compressed CMB distance priors and one supernova sample at a time: Pantheon+, Union3, DES-Dovekie. $\Delta\chi^2$ is relative to ΛCDM: negative is better. σ8 is the amplitude of the matter fluctuations today on 8 h^{-1} Mpc.

what	result

MM 2.32 tested with three registered readings, all with $w \geq -1$	"in tension" overall, 2.1–3.1σ (details in section 4)
parametric reconstructions without MM, with ΛCDM-type early physics	in the parametrisations tried, the best-localised deviations are at $z \approx 0.4$–1; no significant extra factor is found on the CMB distance
the clock idea, with $S = [D/D0]^n$ and the ΛCDM growth curve D	$\Delta\chi^2 = -6.6$ / −11.6 / −10.3; CPL has a χ^2 lower by 0.6–2.3, with one more parameter
the clock follows the growth that its own history produces (no external growth curve)	$\Delta\chi^2 = -7.0$ / −11.9 / −10.6; CPL has a χ^2 lower by 0.3–2.0
growth predicted by that self-consistent version, with Planck's amplitude	$\sigma8 = 0.797$–0.808, compatible with the growth of SDSS, DESI and 6dFGS; χ^2 close to that of ΛCDM
effective equation of state of the clock histories	$w \approx -0.8$ today, crosses −1 between $z \approx 0$ and 0.6, the shape favoured by DESI's w0wa fits
growth of structures for the MM 2.32 family	with Planck's primordial amplitude fixed, compatible in most of the cases tried; not distinguished from ΛCDM by these measurements; median $\sigma8 = 0.790$ along the chain (exceptions and interval: section 8)

How to read the main rows.

- The published MM 2.32 always has $w \geq -1$. With today's data it is *in tension*, at 2.1–3.1σ.
- The clock idea does better than ΛCDM: $\Delta\chi^2$ from −6.6 to −11.6, depending on the supernova sample. CPL does a little better still, but it has one more parameter.
- The version in which the clock follows its own growth does better still: $\Delta\chi^2$ from −7.0 to −11.9.
- The growth predicted by this version agrees with the growth measurements.

What is fitted or assumed.

- The law that links growth and clock is *assumed*. The model is *phenomenological*, that is, chosen to describe the data. It is not derived from the principles of MM.
- The exponent $n \approx 3.6$–3.7 is fitted to the data, not derived.
- The amplitude δ0 is fixed by today's normalisation. It is not predicted.
- That distances follow the read rate is one more assumption. Changing only the clock would not change the measured distances.
- The growth data and the expansion data are partly tied to each other.
- With the same cosmological ingredients and the same dynamics of the perturbations, these data cannot tell the clock reading from a substance with the same expansion history.
- The study is exploratory and preliminary.

Why it is proposed. A single rule answers several requirements together: acceleration without Λ, a late onset, the shape with a changing w, and a growth compatible with the data. It is coherence, not proof (section 10).

Where to look. The hypothesis is in section 7. The test of MM 2.32 is in section 4. The reconstruction from the data alone is in section 6. The growth of the 2.32 family is in section 8.

Labels used in this deposit

The deposit uses many abbreviations. This table explains them. You can come back to it when you meet one.

label	meaning
T1	test of the published expansion history of MM 2.32 against DESI DR2 (section 4)
L1, L2, L3	the three registered readings of 2.32 in T1: CPL restricted to $w \geq -1$ at all times; the registered "thawing" edge of CPL, $w0 + wa = -1$ with $w0 \geq -1$, that is $w(a) = -1 + (1 + w0)\,a$; the 2.32 model itself within its parameter ranges

T2–T7	further translations of the author's working premises into expansion histories (section 5)
R1	reconstruction of what the data ask, with no MM input (section 6)
T8, T8a–T8g	the steps of the clock hypothesis (section 7): T8a what the data ask without Λ; T8 three tracers; T8c growth; T8d–T8e tracer from measured growth; T8f check with computed growth; T8g self-consistent version
K1, K2, K3	tracers S(z): K1 collapsed mass fraction; K2 ΛCDM linear growth raised to a power n; K3 growth of a universe without dark energy (S = a^n)
G0, B1–B3	growth of structures for the 2.32 family: G0 for histories selected from T1–T7; B1 check of the coupled MM Boltzmann code; B2 noise of the readings; B3 growth along the 2.32 chain (section 8)
C1, C2, C3	data combinations: C1 = DESI BAO + CMB priors; C2 = supernovae + CMB priors; C3 = all three (the main one)
PP, U3, DV, D5	supernova samples: Pantheon+, Union3, DES-Dovekie; DES-SN5YR as a check only
CHW	compressed CMB distance priors of Chen, Huang and Wang (2019)
S1, S2, S3	growth data sets: S1 SDSS; S2 DESI DR1 full-shape and peculiar velocities; S3 6dFGS
holds / in tension / disfavoured / excluded	registered decision zones, usually $\Delta\chi^2$ relative to ΛCDM below 4 / 4–9 / 9–25 / above 25; labels of a criterion, not probabilities. T1 uses its own rule based on simulations (section 4)
fast code / full check	the fast code does the fits; the full check re-evaluates the fitted points with CAMB 2.0.4 and HyRec-2

Two rows to remember.

- **The decision zones.** They are fixed before the calculations. Usually: $\Delta\chi^2$ below 4 relative to ΛCDM, the hypothesis *holds*; between 4 and 9 it is *in tension*; between 9 and 25 it is *disfavoured*; above 25 it is *excluded*. They are the labels of a criterion, not probabilities.
- **The two codes.** The *fast code* does the fits. The *full check* redoes the calculations at the points found, with more precise programs.

1. What this deposit is, and what it is not

What it is. It is *one link* of the MM programme: the one on dark energy.

What it is not.

- It is not a derivation of dark energy from first principles.
- It does not show that the clock reading is physically different from a dark energy with the same expansion history. With the ingredients and the perturbation rule adopted here, the two give the same growth (section 7.4).

How the work was done, in short. Each test was registered before its calculation. The registrations were written knowing the results of the earlier tests on the same data; each one says what was already known. The additions decided after seeing the results are labelled as such. The failed predictions are reported as failed. Each block was checked by a second AI assistant. The details are at the end, in *How the work was done*.

2. Prior work first

These are the published results that the work uses.

topic	published result used here
BAO and evolving dark energy	DESI Collaboration, DESI DR2 Results II, arXiv:2503.14738 (2025): in its combinations with the CMB and supernovae, w0waCDM is preferred over ΛCDM at 2.8–4.2σ, with $w_0 > -1$ and $w_a < 0$

growth from full-shape clustering	DESI Collaboration, DESI 2024 V, arXiv:2411.12021; DESI peculiar velocities, arXiv:2512.03230
SDSS BAO and growth	eBOSS Collaboration, Alam et al., PRD 103, 083533 (2021); BOSS DR12, Alam et al., MNRAS 470, 2617 (2017); 6dFGS, Beutler et al., arXiv:1204.4725
supernovae	Pantheon+, Brout et al., ApJ 938, 110 (2022); Union3, Rubin et al., arXiv:2311.12098; DES-Dovekie, Popovic et al., arXiv:2511.07517, which reports a reduced preference for evolving dark energy compared with DES-SN5YR
CMB distance priors	Chen, Huang, Wang, JCAP 02, 028 (2019), arXiv:1808.05724, derived in ΛCDM from Planck 2018 TT,TE,EE+lowE without the lensing-reconstruction likelihood; Planck 2018 VI, arXiv:1807.06209
weak lensing S8	KiDS-Legacy, arXiv:2503.19441; DES Y3, arXiv:2105.13549; HSC Y3, arXiv:2304.00701
parametrisations	Chevallier, Polarski, IJMPD 10, 213 (2001); Linder, PRL 90, 091301 (2003) (CPL); "thawing" and "freezing" fields, Caldwell, Linder, PRL 95, 141301 (2005)
crossing w = −1	Vikman, PRD 71, 023515 (2005); Hu, PRD 71, 047301 (2005); PPF treatment, Fang, Hu, Lewis, PRD 78, 087303 (2008)
clocks of observers in structures	timescape cosmology, Wiltshire, NJP 9, 377 (2007) and arXiv:0909.0749 (2009); Seifert et al., MNRAS Letters 537, L55 (2025), who report with Pantheon+ ln B > 5 for timescape over ΛCDM on the full sample and ln B > 1 with z > 0.075 (a result of that comparison, not of the model here)
backreaction	Buchert, Gen. Rel. Grav. 32, 105 (2000), arXiv:gr-qc/9906015: the averaged equations of an inhomogeneous universe contain terms absent in Friedmann's
collapsed fraction, transfer function	Press, Schechter, ApJ 187, 425 (1974); Eisenstein, Hu, ApJ 496, 605 (1998)
codes	CAMB, Lewis, Challinor, Lasenby, ApJ 538, 473 (2000) (version 2.0.4); HyRec-2, Lee, Ali-Haïmoud, PRD 102, 083517 (2020); CLASS, Blas, Lesgourgues, Tram, JCAP 07, 034 (2011), in the coupled MM version of the 2.32 kit

How to read the first row. DESI, combining its data with CMB and supernovae, prefers an evolving dark energy, at 2.8–4.2σ. Today's value, w0, is above −1, and wa is negative.

The closest precedent is *timescape* cosmology.

- Timescape attributes the apparent acceleration to a difference between clocks. Observers inside bound structures, like galaxies, have a clock different from the mean clock over the whole volume.
- Timescape uses walls, voids and Buchert's averaging.
- The model of this work uses neither voids nor averages. It is a relation with a single function: it links the expansion rate read by the observer to a tracer of the growth of structures.

3. Data and procedure

The data. The combinations C1–C3 and the samples are in the label table. Third-party data are not redistributed.

The codes.

- A *fast code* does the fits. It computes the background with closed formulas and uses a tabulated recombination.
- A *full check* redoes the calculations at the minima with the programs CAMB 2.0.4 and HyRec-2. The dark sector is passed as a w(a) table, through PPF.
- **Where the full check fails:** for K1, T8d and two variants of R1. It is declared in each note.
- **Where it passes with difficulty:** T8e and T8g pass it only with the tolerance on the density widened to 10^{-4}.

Approximations to keep in mind.

- The CHW priors are derived within ΛCDM. Using them for other histories of the universe is an approximation.
- The three supernova samples use the same BAO and CMB data. So they are not three independent confirmations.
- The growth data are never summed with each other.

4. MM 2.32 against DESI DR2 (T1)

What was tested. The history of the universe published in 2.32 has $w \geq -1$ at every epoch, by construction. Three readings were registered: L1, L2 and L3 (see the label table).

How.

- For L1 and L2 an analytic test statistic was used, checked with simulations.
- For L3 a calibration with simulations was used.
- In total: 22 runs of 400 simulations, four extended to 2000, and 3728 tail and control simulations refitted separately.
- The intervals are computed with the Clopper-Pearson method.

The result (C3).

reading	Pantheon+	Union3	Dovekie	overall
L1	2.15σ, in tension	3.14σ, disfavoured	2.55σ, in tension	**in tension**
L2	2.08σ	3.10σ	2.50σ	**in tension**
L3 (first 400 simulations)	2.38σ	2.81σ	3.02σ	**in tension**

How to read it. All three readings turn out *in tension* overall. With Union3, L1 reaches 3.14σ and is *disfavoured*.

- No reading meets the registered criterion "falls with today's data".
- The zones of L3 for Union3 and Dovekie are uncertain, because the simulations are finite in number. And the MM calibration uses a single null point.
- Without the DESI BAO (combination C2), L1 and L2 hold. L3 is not calibrated there.

A correction to the minimum of 2.32.

- With more starting points one finds $\chi^2 = 1419.66$ instead of 1420.02.
- This minimum lies at the edge of the allowed parameter range ($u_c = -3$, $\delta u = 0.1$).
- The improvement over ΛCDM is 1.52, not 1.17.
- It is not a measured preference for a late and narrow transition. Widening the range, the minimum follows the new edge.

5. Other readings of the author's premises (T2–T7)

Each one is a registered translation of a working premise into an expansion history. **A negative result concerns that translation, not the premise.**

test	translation	result
T2	dark energy as a distant effect of the splitting of the relational network (four laws D1–D4)	D3 is equal to ΛCDM by construction; the laws that add an expansion rate are disfavoured or excluded. An effect on supernova fluxes only, on top of ΛCDM (D4), recovers about a third of the w0wa improvement; without dark energy (D4b) it fits very badly
T3	the value of the dark-energy density from twelve "rate × epoch" combinations	none gives $\Omega_\Lambda \approx 0.698$; a relation with $3\sqrt{3}$ returns Ω_Λ within +0.0014/+0.0022, using flatness and the fitted parameters, but does not fix the absolute density
T4	the rate from when objects come into being and from the distance assigned to them in the tested translation ("diffraction distance"; see the final note of T4)	none of the 18 combinations within 0.02 of $\Omega_\Lambda = 0.698$ (central values; not a full statistical test)
T5	dark energy as a fraction of the network passed on locally, switched on at redshift z_v (three shapes)	a late switch-on of Λ holds from $z_v = 1.5$ on the registered grid (best $\Delta\chi^2 = -3.9$ at $z_v = 2.5$, still above CPL by +3.7/+10.0/+6.9). The shapes tied to the collapsed fraction or to growth as a density are excluded, except one case with the collapsed fraction, which is disfavoured. The shape with the collapsed fraction has no full check

T6, T7	a stretching of the read distances, largest on the CMB	$\Delta\chi^2$ from −4.6 to −5.1, a part of the CPL gain. The author later withdrew the premise: section 6 puts the deviation at z ≈ 0.5–1, not on the CMB

What the table says, row by row.

- **T2.** Dark energy as a *distal* effect, that is, a distant one, of the *splitting* of the network of relations. Four laws were tried, D1–D4. D3 is equal to ΛCDM by construction. The laws that add an expansion rate are *readings that fall*: disfavoured or excluded. An effect only on the light of supernovae, on top of ΛCDM (D4), recovers about a third of the advantage of w0wa. Without dark energy (D4b) it does very badly.
- **T3.** The value of the dark-energy density was searched for in twelve "rate × epoch" combinations. None gives $\Omega_\Lambda \approx 0.698$. A relation with $3\sqrt{3}$ gives a close Ω_Λ, within +0.0014/+0.0022. But it uses flatness and the fitted parameters, and it does not fix the absolute density.
- **T4.** The rate is obtained from when objects come into being and from the distance assigned to them in the tested translation ("diffraction distance"). None of the 18 combinations gets within 0.02 of $\Omega_\Lambda = 0.698$. They are central values, not a full statistical test.
- **T5.** Dark energy as a part of the network passed on locally, which switches on at redshift z_v. A late switch-on of Λ holds from $z_v = 1.5$ onwards. The best $\Delta\chi^2$ is −3.9 at $z_v = 2.5$, but it stays above CPL by +3.7/+10.0/+6.9. The shapes tied to the collapsed fraction or to growth as a density are excluded, except one case with the collapsed fraction, which is disfavoured. The shape with the collapsed fraction has no full check.
- **T6 and T7.** A stretching of the read distances, largest on the CMB. It gives $\Delta\chi^2$ from −4.6 to −5.1: a part of the gain of CPL. **The author later withdrew this premise.** Section 6 in fact shows that the deviation is at z ≈ 0.5–1, not on the CMB.

6. What the data ask, without MM (R1)

Here MM does not enter. We ask the data where they want a deviation from ΛCDM. *Reconstructions* are made: the history of the universe is left free in some redshift bands, with the early-universe physics of ΛCDM.

reconstruction (extra parameters)	$\Delta\chi^2$ relative to ΛCDM (PP / U3 / DV)	relative to CPL
one distance factor $D(1 + \varepsilon(z))$, nodes 0.5, 1, 2 and the CMB (4)	−9.1 / −9.2 / −9.3	−1.5 / +4.6 / +1.5
radial and transverse distances separate (7)	−13.1 / −14.1 / −13.3	−5.5 / −0.3 / −2.5
dark-energy density bounded in bands (6; fast code only)	−13.8 / −15.5 / −12.7	−6.2 / −1.6 / −1.9

Where the deviations are.

- In the parametrisations tried, the best-localised deviations are at z ≈ 0.4–1.
- There is no significant extra factor on the CMB distance ($\sqrt{\Delta\chi^2}$ 0.2–0.9).
- The dark-energy density is above today's at z ≈ 0.4 (+6/+14 %) and at z ≈ 0.8 (+16/+26 %). At the null value $\sqrt{\Delta\chi^2}$ is 1.3–2.2. These are local indicators, not global significances.
- Beyond z ≈ 1.2 the density is not determined.

Radial and transverse distances. The *radial* distance is along the line of sight; the *transverse* one is across it.

- The radial one asks for −3.6/−4.5 % at z ≈ 0.5.
- The transverse one asks for −1.3/−2.1 % at z ≈ 1.
- Their difference is below 1.8 times its error at every node.
- The reconstruction with the dark-energy density bounded in bands does just as well with one parameter fewer. It is evaluated only with the fast code.
- So these data do not require separate readings for the radial and transverse distances.

7. The hypothesis: the read expansion rate follows the growth of structures (T8)

7.1 The translation (one among several possible)

- **Bare universe.** Matter and radiation (with neutrinos), and no dark energy. Its expansion rate is $H_{bare}(z) = 100\sqrt{G(z)}$ km/s/Mpc, where G is the physical density in ω units.

- **Read rate.** H_read(z) = H_bare(z) [1 + δ0 S(z)], with S(0) = 1 and S → 0 without structures.
- **δ0.** It is fixed by the *flatness* of the universe: (1 + δ0)² = h²/G(0). It is a normalisation, not a prediction of the amplitude.
- **The distances.** D_M = ∫ c dz / H_read. **This is one more physical assumption.** For this reason the model is phenomenological, not a derivation from the premises on the clock.
- **The sign.**
- With this form the *deceleration parameter* is q = q_bare − δ0 S′/(1 + δ0 S), where S′ = dS/d ln a ≥ 0. So the growth of structures can produce acceleration.
- For δ0 > 0, the opposite form, H_bare/(1 + δ0 S), decelerates more than the bare universe. With the same δ0 it does not preserve today's normalisation. It is used only to check the sign.
- The author's premise «il tempo proprio rallenta dove la struttura cresce» ("proper time slows where structure grows") is read accordingly: the slowed proper time makes the measured rates appear faster.
- **The equivalence.** The same H(z) can be written as an effective dark density G[(1 + δ0 S)² − 1]. Standard cosmology would read it as a substance.

7.2 What the data ask without Λ (T8a)

- From the minima of ΛCDM, of CPL and of the band reconstruction comes H_bare(0) = 37.5–37.8 km/s/Mpc.
- The read rate must exceed the bare one:
- by 75–82 % today;
- by 30–32 % at z = 0.5;
- by 12–14 % at z = 1;
- by 3–4 % at z = 2;
- by about 0 at z = 5.
- So S(z) is 0.36–0.42 at z = 0.5 and 0.15–0.17 at z = 1.

7.3 Three tracers of structures (C3)

tracer	registered grid: Δχ² relative to ΛCDM (PP / U3 / DV)	zone	best after the declared refinements	relative to CPL
K2: S = (D/D0)^n, D = ΛCDM linear growth	n = 3.5: −3.95 / −11.60 / −8.07	holds	n = 3.7 / 3.5 / 3.6: −6.56 / −11.60 / −10.27	+1.04 / +2.25 / +0.57
K3: S = a^n	n = 2.5: +13.0 / +12.3 / +13.7	disfavoured	unchanged	+20.6 / +26.2 / +24.5
K1: collapsed fraction (Press–Schechter; fast code only)	10^{13} M_⊙/h: +119 / +22 / +133	excluded	+39 / +2.8 / +30	—

How to read it.

- **K2** uses the linear growth of ΛCDM raised to the power n. It *holds*: after the refinements, Δχ² from −6.56 to −11.60. It stays a little above CPL.
- **K3** uses S = a^n, a growth without dark energy. It is *disfavoured*: positive Δχ², that is, worse than ΛCDM.
- **K1** uses the *collapsed fraction*: the part of matter already gathered into bound objects. It is *excluded*. It is computed only with the fast code.

K2 at its minima.

- δ0 = 0.769–0.795, h = 0.666–0.677, Ω_m = 0.310–0.319.
- n lies within 3.40–3.76 at Δχ² = 1.
- The full check reproduces K2 and K3 within 0.004 in χ².
- **The effective history (K2):** w_eff = −0.83/−0.78/−0.80 today. It crosses −1 at z = 0.32/0.49/0.40. It is about −1.2 at z = 3.

7.4 Growth with the read history (T8c, K2 at n = 3.5)

- **Growth that follows H_read** (CAMB-PPF, with Planck's A_s and n_s). It is compatible with S1, S2 and S3. σ8 = 0.793–0.794, S8 = 0.815–0.819. The read age of the universe is 13.75 Gyr (billions of years).
- **Growth of the bare universe.** It is incompatible even with a free amplitude (S2: χ² 36.0 for 7 points). The bare age is 17.3 Gyr.
- **Equivalence.** With the same ingredients and this perturbation rule, the growth coincides with that of a history

with the same H(z). So the expansion and growth data alone do not tell the clock reading from a substance.

7.5 Tracer built from the measured growth, and a check (T8d, T8e, T8f)

What was done.

- Instead of the ΛCDM growth curve, a curve fitted to the measured fσ8 (S1 or S2) is used.
- Growth must not be faster than in the matter era: f ≤ 1. It is an assumption of the tested family.
- Expansion and growth are fitted together, as if they were independent. The joint covariance of DESI growth and DESI BAO is not treated.

The result (T8e).

- It does better than ΛCDM plus a free growth curve by 8.7 / 12.7 / 10.9 with DESI growth (S2).
- It does better by 8.2 / 11.4 / 9.8 with SDSS (S1).
- It has as many parameters as CPL plus the curve.
- These results are preliminary.
- σ8 today sits at the limit set by f ≤ 1.

The check (T8f).

- The CAMB prediction was rescaled to the fitted σ8(0).
- For DESI growth with Pantheon+ and with Dovekie, the shape of the growth passes both registered checks, at the tested redshifts.
- Union3 slightly exceeds the pointwise limit: 1.07 for a limit of 1.
- Separately, the fitted present-day amplitude differs by less than 0.75 % from the CAMB prediction with the adopted Planck A_s. It is not a full statistical test.
- **With SDSS growth the check does not pass.**
- The details, including T8d and the sensitivity to the growth curve, are in the final note of the block.

7.6 Self-consistent version: the clock follows its own growth (T8g)

The definition.

- S = (D/D0)^n, where D is the linear growth computed in the same read history.
- It is a *fixed point*: the clock depends on the growth, and the growth depends on the clock. It is found by iteration, that is, by redoing the calculation until the result no longer changes.
- There is no fitted growth curve and no growth data in the fit.
- Parameters: h, ω_b, Ω_m and n.

The expansion (C3).

	Pantheon+	Union3	Dovekie
Δχ² relative to ΛCDM	−7.03	−11.85	−10.59
Δχ² relative to CPL	+0.57	+2.00	+0.25
n	3.707	3.575	3.692
full check − fast code, evaluated at the minimum of the fast code	+0.0042	+0.0045	+0.0040

Zone: *holds*. These are the best points found starting from three different points. It is not shown that the fixed point is unique, nor that the minimum is the global one.

Growth as a test. CAMB-PPF is used in the read history, with the central values of Planck's A_s and n_s, without any free amplitude.

- σ8 = 0.797–0.808 and S8 = 0.820–0.824.
- The χ² is 6.26–6.80 for SDSS (ΛCDM 6.6), 3.77–4.10 for DESI (4.0) and 0.03–0.05 for 6dFGS (0.02). It is

compatible in all the sets.
- A free amplitude would be within 1.51σ of 1.
- The growth comparison is still partly tied to the expansion data.

Internal coherence.

- The normalised growth D(z)/D(0) of the fixed point agrees with the CAMB growth (cold matter plus baryons) within $2 \cdot 10^{-4}$.
- The check is done at seven redshifts between 0 and 2, at k = 0.05 Mpc^{-1}.
- It does not check other scales, the absolute amplitude or f.
- The CAMB curve is not in turn reinserted into the clock law.

The effective history. w_eff = −0.83/−0.80/−0.83 today and −1.04/−1.00/−1.04 at z = 0.5. So it crosses −1 between z = 0 and 0.5.

The limits. The clock law, the distance rule and the linear growth, equal at all scales, are assumptions. And there is not yet any prediction that only the clock can make.

8. Growth of structures for the MM 2.32 family (G0 and part B)

G0: histories selected from T1–T7, with Planck's amplitude.

- They are all compatible with the fσ8 of SDSS, DESI DR1 and 6dFGS according to the registered rule.
- There is one exception: a case in tension at 2.1σ (thawing with Union3, with the full SDSS likelihood).
- The measurements do not distinguish them from ΛCDM ($|\Delta\chi^2| \leq 0.96$).

B1: the coupled MM Boltzmann code. A *Boltzmann code* computes how perturbations grow in the universe.

- At three points of the chain, its growth normalised to today agrees with CAMB on the same background, within the registered thresholds.
- The background agrees within 2.2×10^{-6}. The MM/ΛCDM growth ratio agrees within 6.7×10^{-4} on the tested grid.
- This comparison does not isolate the perturbations of the dark sector. And it does not verify the σ8 normalisation used in B3.

B3: 400 samples of the published chain of 2.32.

- Median σ8 = 0.790 (95 %: 0.771–0.800).
- It is below ΛCDM with the same parameters in every sample (median −0.86 %).
- No paired $\Delta\chi^2$ exceeds 4.
- 2 % of the samples are in tension in the SDSS growth, only through the amplitude (at most 2.44σ).

B2: the noise of the readings. The record counts have a *noise*: small random oscillations. It was checked how much they disturb matter.

- At the tested points, the root-mean-square response of matter to this noise (rms: the square root of the mean of the squares, which measures how large the fluctuation is) is bounded, for each Fourier mode (each wave into which the density is broken down).
- The bound is 1.1×10^{-5} rms(δ_m) for the density and 3.8×10^{-5} f rms(δ_m) for $\delta' = d\delta/d \ln a$.
- These bounds use the adopted source model and normalisations, in the branch with no additional vacuum component. They hold in linear theory, below the horizon, without back-reaction or relativistic terms.
- In this approximation the noise is negligible (registered threshold 10^{-3}).
- It is not a bound on fσ8.

The scope.

- The histories are selected, without MM-specific perturbations in G0.
- A_s is fixed to Planck's central value.
- The growth data are not all independent of the expansion data.

9. How MM 2.32 should be read now

- **The translation, not the premises.** 2.32 translated the record counts into an energy density and a pressure. There it was declared as an imported effective postulate. The constraint $w \geq -1$ comes from that translation, not from the author's premises.
- **The published version stays as it was computed.** Its evaluation is that of section 4: "in tension".
- **Still to do.** A translation of the counts of 2.32 into the clock reading of section 7 has not yet been written. Renaming the same H(z) as a "rate" would not change its w.

10. What the hypothesis answers, and what is open

The hypothesis answers several requirements together. It is coherence, not a set of independent confirmations.

1. Late acceleration without Λ, with good fits to the DESI DR2 BAO, to the CMB distance priors and to three supernova samples.
2. An effective $w \approx -0.8$ today, which crosses -1 at $z \approx 0$–0.6. It is in the range favoured by these parametrisations and by these data.
3. A late onset, because the tracer follows the growth of structures. How late depends on the chosen form and on the fitted n; and structures grow earlier too. The reconstructions without MM place the best-constrained deviations at $z \approx 0.4$–1, on the same data: a qualitative coherence.
4. A self-consistent version whose growth, with Planck's amplitude, is compatible with the measurements.
5. A background and an early dark fraction under control. Not all the physics of the early universe is controlled, nor the CMB spectrum.
6. Two working premises of the author, read as one possible interpretation: proper time tied to the local structure, and space that changes only through time. The code, however, uses a global cosmological curve, not the local structure or the environment.

The author's motivation. The clock is tied to a quantity that exists and that is observed through growth (here not statistically independent of all the expansion data). A substance with the same H(z), instead, would have to be postulated for the purpose. This is the reason why the author studies the hypothesis. It is not a result of the χ^2 values. The observed growth does not on its own support the particular law that ties it to the clock.

What is open.

- The exponent n is fitted: 3.5–3.7 for K2, 3.575–3.707 for T8g, up to 3.93 for T8e with SDSS.
- The law that links growth and clock, and the distance rule, are assumptions.
- The joint covariance of DESI growth and DESI BAO is not treated.
- Growth enters the clock law as a single linear factor, equal at all scales.
- There is not yet any clock-specific prediction, for example a dependence on the environment.
- The translation of the counts of 2.32 into the clock reading is not yet written.

How the work was done

The registrations. Each test was registered before its calculation: the translation, the data, the decision zones and the author's predictions. Each registration has a SHA-256 fingerprint and a UTC time in a REGISTRO.txt. The registrations were written knowing the results of the earlier tests on the same data; each one says what was known.

Additions and failures. The additions decided after seeing the results are labelled as such. The failed predictions are reported as failed.

The independent check. Each block was checked by gpt-6-astra via Codex, in a number of rounds from one to nine. Its reports and the corrections are in the records.

The final notes. Each block has a final note (NOTA_ESITO_*.md, in Italian) with every number and limit:

- T1: blocchi/01_T1_tensione_DESI/NOTA_ESITO_T1.md;
- T2–T7: from blocchi/02_… to blocchi/06_…;
- R1: NOTA_ESITO_R1.md;
- T8: NOTA_ESITO_T8.md and NOTA_ESITO_T8g.md;
- G0 and part B: NOTA_ESITO_G0.md and NOTA_ESITO_B.md, with the full list of limits and the two earlier numerical versions of B2, which are not used;
- the reading of 2.32: blocchi/09_lettura_della_2.32.

The data. The origins of the third-party data and their SHA-256 fingerprints are in FONTI_DATI.md.

How to redo the calculations.

- **blocchi/** contains each block with its scripts, results, registrations, notes, REGISTRO.txt, and the prompts and reports of the checker. The working files and the folder names are in Italian.
- **dipendenze/mm_core** contains the Python modules of the MM 2.32 fitting core used here. For example distance_priors.py, with the CHW means, the symmetrised errors and the correlation matrix. The native components HyRec and coupled CLASS are in the MM 2.32 kit.
- **COLLAUDO.md** reports a re-evaluation of the main numbers from the saved minima, on the author's machine, with the code of the working folders. That code is byte-for-byte identical to the code in blocchi/. The scripts contain absolute paths of that machine, to be adapted.
- **Software:** Python 3.13, NumPy, SciPy, CAMB 2.0.4, Astropy (for the Union3 FITS files).
- **IMPRONTE.sha256** lists the SHA-256 of every file of the deposit, except itself.

The records.

- In each block there are:
- REGISTRAZIONE_* (registrations), CORREZIONI_* (corrections after each check round), AGGIUNTA_* (additions);
- BOZZA_* (drafts), NOTA_ESITO_* (final notes) and REGISTRO.txt (time-stamped registry);
- astra_*/PROMPT_*.md and RAPPORTO.md (the prompts and reports of the checker).
- The blocks are:
- 01 T1 (nine check rounds);
- 02–05 T2–T5;
- 06 T6–T7 (premise withdrawn);
- 07 R1;
- 08 T8 (T8a–T8g);
- 09 reading of MM 2.32;
- 10 G0;
- 11 part B.
- The checks of the archived text are in astra_deposito_*.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **b.** A free amplitude that multiplies the predicted fσ8. It serves to test the shape of growth separately.
- **BAO.** Baryon acoustic oscillations: a typical distance in the distribution of galaxies, used as a cosmic ruler.
- **Bare universe.** Matter and radiation (with neutrinos), no dark energy: H_bare = 100 √G.
- **CMB.** The cosmic microwave background, the oldest light of the universe.
- **CPL (w0wa).** The dark-energy equation of state w(a) = w0 + wa (1 − a).
- **Dark energy.** The name given to what makes the expansion of the universe accelerate.
- **Decision zones.** The thresholds fixed before the calculations: holds, in tension, disfavoured, excluded.
- **Δχ².** The difference in χ^2 from ΛCDM at the best points. Negative means a better fit.
- **fσ8.** The growth rate times the amplitude of matter fluctuations on 8 h^{-1} Mpc.
- **Growth of structures.** The gathering of matter into galaxies and clusters, through gravity.
- **ΛCDM.** The standard model of cosmology, with a cosmological constant.
- **MCD (in the author's language).** A local structure of the relational network. Here it is represented only through the global growth of structures.
- **Phenomenological.** Chosen to describe the data, not derived from a theory.
- **PPF.** A treatment of the dark sector that allows w to cross −1 in the CAMB program.
- **Read rate.** The expansion rate as read by the observer's clock: H_read = H_bare (1 + δ0 S).
- **Redshift (z).** How much the light has been stretched during its journey.
- **Registration.** A test written and dated before the calculations.
- **S8.** S8 = σ8 (Ω_m/0.3)^0.5.
- **σ8.** How lumpy matter is today, on spheres of 8 h^{-1} Mpc.
- **Tracer S(z).** A function of the growth of structures: 1 today and 0 without structures.

- **w.** The pressure divided by the energy density. For the cosmological constant it is −1.

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